Agent-Based Model of Speed-Transfer-Mediated Collective Rhythmicity — Equations and Parameters

1 Overview

The simulation models $N_1 + N_2$ agents on a two-dimensional rectangular arena representing the comb surface. Group 1 ("rhythmic" / foragers) receives an external sinusoidal speed driver; Group 2 ("non-rhythmic" / hive bees) does not. Speed transfer occurs when agents interact within a spatial proximity threshold, allowing the rhythmic pattern to propagate from Group 1 to Group 2.

2 State Variables

Each agent i at time step t is described by:

- Position $\mathbf{x}_i^{(t)} = (x_i, y_i)^{(t)} \in \mathbb{R}^2$
- Heading $\theta_i^{(t)} \in [0, 2\pi)$
- Accumulated speed transfer $\Delta s_i^{(t)} \geq 0$

3 Initialization ($t = 0$)

Positions are drawn from group-specific bivariate Gaussians centred at μ_g with covariance $\Sigma_g = \sigma_s \mathbf{I}_2$:

$$\mathbf{x}_i^{(0)} \sim \mathcal{N}\big(\mu_g,\, \Sigma_g\big), \qquad \theta_i^{(0)} \sim \text{Uniform}(0,\, 2\pi), \qquad \Delta s_i^{(0)} = 0. \tag{1}$$

4 Per-Step Update ($t \to t + 1$)

At every time step t (with integer time label $\tau = t + 1$), agents are updated in the following order.

4.1 External Driver (Group 1 only)

Group 1 receives a sinusoidal speed boost that peaks once per simulated day of duration T_{day}:

$$e^{(\tau)} = \frac{\sigma_s\, \mu_1^{\text{fwd}}}{2} \left(\sin\!\left(\frac{2\pi\, \tau}{T_{\text{day}}} \right) + 1 \right), \tag{2}$$

where σ_s is the speed scale and μ_1^{fwd} is Group 1's mean forward speed. Group 2 receives $e^{(\tau)} = 0$.

4.2 Forward Speed Sampling

Forward speeds are drawn from a Gamma distribution parameterised by the group-specific mean μ_g^{fwd} and standard deviation σ_g^{fwd}, scaled by σ_s:

$$v_i^{\text{raw}} \sim \text{Gamma}\left(k = \frac{(\sigma_s\,\mu_g^{\text{fwd}})^2}{(\sigma_s\,\sigma_g^{\text{fwd}})^2},\ \theta = \frac{(\sigma_s\,\sigma_g^{\text{fwd}})^2}{\sigma_s\,\mu_g^{\text{fwd}}}\right), \qquad v_i^{\text{fwd}} = \max\bigl(v_{\min},\ v_i^{\text{raw}}\bigr), \tag{3}$$

where $v_{\min}$ is a per-group lower-bound on forward speed and (k, θ) are the shape and scale of the Gamma distribution. The total instantaneous speed is:

$$s_i^{(\tau)} = v_i^{\text{fwd}} + e_i^{(\tau)} + \Delta s_i^{(\tau)}. \tag{4}$$

4.3 Turning Noise

Heading changes are drawn from a Von Mises distribution:

$$\delta\theta_i \sim \text{VonMises}\left(\mu = \mu_g^{\text{turn}},\ \kappa = \frac{1}{(\sigma_g^{\text{turn}})^2}\right). \tag{5}$$

4.4 Velocity and Position Update

The velocity vector combines the speed with the current heading:

$$\mathbf{v}_i = s_i^{(\tau)} \begin{pmatrix} \cos\theta_i^{(t)} \\ \sin\theta_i^{(t)} \end{pmatrix}. \tag{6}$$

Homing pull. Each group is attracted toward its home location μ_g:

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \sigma_s\,h\,\mu_g^{\text{fwd}}\,\frac{\mu_g - \mathbf{x}_i^{(t)}}{\left\|\mu_g - \mathbf{x}_i^{(t)}\right\|}, \tag{7}$$

where h is the homing pull coefficient.

The state is then updated:

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \mathbf{v}_i, \tag{8}$$

$$\theta_i^{(t+1)} = \theta_i^{(t)} + \delta\theta_i. \tag{9}$$

4.5 Wall Reflection

The arena is bounded by a rectangle $[-L_x/2,\, L_x/2] \times [-L_y/2,\, L_y/2]$. When an agent crosses a boundary, its position is clamped to the boundary and its heading is redirected toward the arena centre:

$$\text{if } x_i^{(t+1)} < -\tfrac{L_x}{2} \text{ or } x_i^{(t+1)} > \tfrac{L_x}{2}: \qquad \theta_i^{(t+1)} \leftarrow \arctan 2\bigl(-y_i^{(t+1)},\, -x_i^{(t+1)}\bigr). \tag{10}$$

An analogous rule applies along the y-axis.

4.6 Spatial Interactions and Speed Transfer

At each time step, agent pairs within Euclidean distance $d_{\text{int}} = \sqrt{\sigma_s}\ \rho$ are identified using spatial hashing. The interaction threshold factor ρ sets the interaction range as a fraction of the characteristic spatial scale $\sqrt{\sigma_s}$. With the default value $\rho = 0.2$, agents interact within $d_{\text{int}} \approx 1.4$ mm, corresponding to physical body contacts between bees on the comb surface.

For each interacting pair (i, j) where $s_j > s_i$, the speed difference is computed and the slower agent receives a boost:

$$\Delta s_i^{(\tau+1)} = \gamma\, \Delta s_i^{(\tau)} + \max_{j \in \mathcal{N}_i} \left(s_j^{(\tau)} - s_i^{(\tau)} \right)^+, \tag{11}$$

where $\gamma \in [0, 1)$ is the exponential decay rate, $\mathcal{N}_i = \{j : \|\mathbf{x}_j - \mathbf{x}_i\| < d_{\text{int}}\}$ is the set of neighbours, and $(\cdot)^+ = \max(\cdot, 0)$.

5 Rhythmicity Analysis

After the simulation, the mean speed across agents within each group is computed for every time step. A sinusoidal model is fitted to these time series by ordinary least squares:

$$\bar{s}_g(\tau) \approx \beta_0 + \beta_1 \sin\left(\frac{2\pi\,\tau}{T_{\text{day}}}\right) + \beta_2 \cos\left(\frac{2\pi\,\tau}{T_{\text{day}}}\right), \tag{12}$$

yielding amplitude $A = \sqrt{\beta_1^2 + \beta_2^2}$ and phase $\phi = \arctan 2(-\beta_2, \beta_1)$. The phase shift between groups is $\Delta\phi = \phi_2 - \phi_1$.

Statistical significance is assessed via a permutation test: the time labels are shuffled n_{perm} times, sine fits are recomputed, and the p-value is the fraction of permuted amplitudes exceeding the observed amplitude.

6 Parameter Table

Table 1: Default parameter values used in all simulations unless otherwise noted.

Symbol	Parameter	Value	Unit
Arena			
L_x	Comb width	354	mm
L_y	Comb height	205	mm
Environment			
σ_s	Speed scale	50	–
h	Homing pull coefficient	0.1	–
γ	Speed transfer decay	0.8	–
ρ	Interaction threshold factor	0.2	–
d_{int}	Interaction threshold	$\sqrt{50} \times 0.2 \approx 1.41$	mm
Timing			
T_{day}	Steps per simulated day	400	steps
T_{sim}	Total simulation duration	1200	steps
Group 1 (rhythmic)			
μ_1	Home position	$(-175, -100)$	mm
μ_1^{fwd}	Mean forward speed	0.05	mm/step
σ_1^{fwd}	Std forward speed	0.02	mm/step
$v_{\min,1}$	Minimum forward speed	-0.05	mm/step
μ_1^{turn}	Mean turning rate	0.0	rad/step
σ_1^{turn}	Std turning rate	0.5	rad/step
Group 2 (non-rhythmic)			
μ_2	Home position	$(0, 0)$	mm
μ_2^{fwd}	Mean forward speed	0.02	mm/step
σ_2^{fwd}	Std forward speed	0.01	mm/step
$v_{\min,2}$	Minimum forward speed	-0.05	mm/step
μ_2^{turn}	Mean turning rate	0.0	rad/step
σ_2^{turn}	Std turning rate	0.5	rad/step
Agent counts			
N_g	Agents per group	$\lfloor 500 \times d \rceil$	–
	(density factor d)	1.0 (default)	–